

2009 Prelim H2 Maths Paper 2

1. Objective: Small angle approximation & Binomial expansions

$$\begin{aligned}
 BC &= BD + DC \\
 &= \frac{h}{\tan \frac{\pi}{6}} + \frac{h}{\tan \left(\frac{\pi}{4} + x \right)} \\
 &= \frac{h}{\frac{1}{\sqrt{3}}} + \frac{h}{\frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x}} \\
 &= h\sqrt{3} + \frac{h(1 - \tan x)}{1 + \tan x} \\
 &\approx h\sqrt{3} + \frac{h(1 - x)}{1 + x} \\
 &= h\sqrt{3} + h(1 - x)(1 + x)^{-1} \\
 &= h\sqrt{3} + h(1 - x) \left[1 + (-1)x + \frac{(-1)(-2)}{2!}x^2 + \dots \right] \\
 &= h\sqrt{3} + h(1 - x)[1 - x + x^2 + \dots] \\
 &= h\sqrt{3} + h(1 - 2x + 2x^2 + \dots) \\
 &\approx h(1 + \sqrt{3} - 2x + 2x^2)
 \end{aligned}$$

2. Objective: Find volume of solid of revolution using substitution

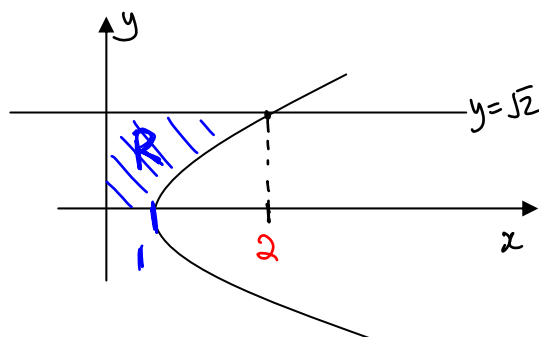
(i) Let $u = \sqrt{x-1} \Rightarrow x = u^2 + 1 \Rightarrow \frac{dx}{du} = 2u$

When $x = 1$, $u = 0$, When $x = 2$, $u = 1$

$$\begin{aligned}
 \int_1^2 x\sqrt{x-1} \, dx &= \int_0^1 (u^2 + 1) \left(\sqrt{u^2 + 1 - 1} \right) (2u) \, du \\
 &= \int_0^1 (u^2 + 1)(u)(2u) \, du \\
 &= \int_0^1 (2u^4 + 2u^2) \, du \\
 &= \left[\frac{2}{5}u^5 + \frac{2}{3}u^3 \right]_0^1 = \frac{16}{15}
 \end{aligned}$$

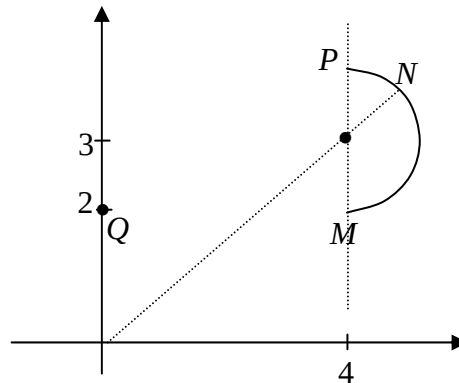
(ii) When $y = \sqrt{2}$, $(\sqrt{2})^2 = x\sqrt{x-1}$

$$\begin{aligned}
 \Rightarrow 2 &= x\sqrt{x-1} \\
 \Rightarrow x^2(x-1) &= 2^2 \\
 \Rightarrow x^3 - x^2 - 4 &= 0 \\
 \Rightarrow x &= 2 \text{ (no other real solutions)}
 \end{aligned}$$



$$\begin{aligned}
 \text{Required volume} &= \text{volume of cylinder} - \pi \int_1^2 y^2 \, dx \\
 &= \pi (\sqrt{2})^2 (2) - \pi \int_1^2 x \sqrt{x-1} \, dx \\
 &= 4\pi - \frac{16}{15}\pi = \frac{44}{15}\pi
 \end{aligned}$$

3. Objective: Loci application



maximum $|z|$ at $N = 6$, minimum $|z|$ at $M = \sqrt{20}$

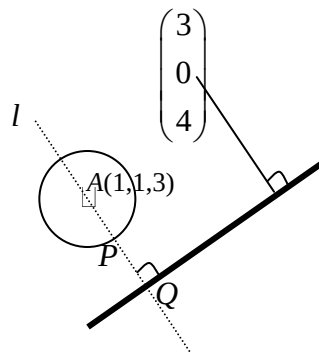
$$\min \arg = 0, \quad \max \arg = \angle PQM = \tan^{-1}\left(\frac{1}{2}\right)$$

4. Objective: Vectors application

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$$

$$\begin{aligned}
 \text{pt } Q: \begin{pmatrix} 1+3\lambda \\ 1 \\ 3+4\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} &= 40 \\
 \rightarrow \lambda &= 1
 \end{aligned}$$

$$\mathbf{q} = \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$$



Let P be pt on sphere nearest to π

$$AQ^2 = 9 + 0 + 16, AQ = 5 \rightarrow AP = 2 \text{ \& } PQ = 3$$

shortest dist between sphere & Plane = 3

$$5\mathbf{p} = 3 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}, \quad \mathbf{p} = \frac{1}{5} \begin{pmatrix} 11 \\ 5 \\ 23 \end{pmatrix}$$

5(a) Objective: Applications of AP & GP

$$\begin{aligned}\text{Sum} &= \sum_{r=1}^{4k} r - \sum_{r=1}^{2k-1} r \\&= \frac{4k}{2}(1+4k) - \frac{2k-1}{2}(1+2k-1) \\&= 2k(1+4k) - k(2k-1) \\&= 6k^2 + 3k \\&= 3k(2k+1)\end{aligned}$$

(i)

Alternatively

$$\begin{aligned}\text{Sum} &= \sum_{r=2k}^{4k} r \\&= \frac{4k - 2k + 1}{2}(2k + 4k) \\&= \frac{2k+1}{2}(6k) \\&= 3k(2k+1)\end{aligned}$$

5(aii) Sum of the odd integers lying between $2k$ and $4k$

$$\begin{aligned}&= \sum_{r=2k}^{4k} r - \sum_{r=k}^{2k} 2r \\&= 3k(2k+1) - 2 \sum_{r=k}^{2k} r \\&= 3k(2k+1) - 2 \left(\frac{k+1}{2} \right) (k+2k) \\&= 3k(2k+1) - (k+1)(3k) \\&= 3k^2\end{aligned}$$

Alternatively

The odd integers lying between $2k$ and $4k$ are $2k+1, 2k+3, 2k+5, \dots, 4k-1$

Let the number of terms be n .

$$4k-1 = (2k+1) + (n-1)(2)$$

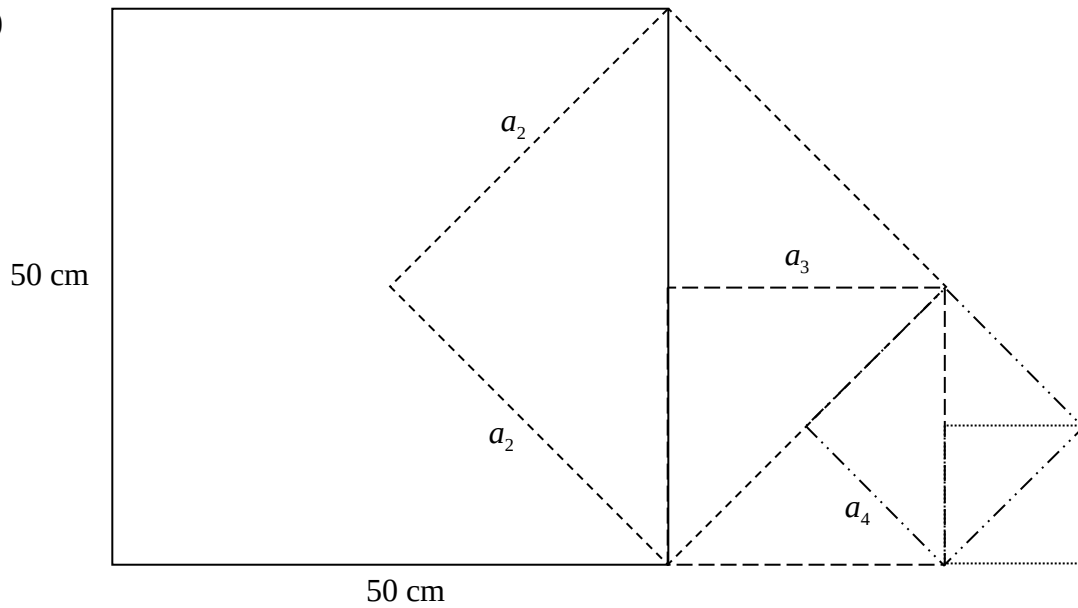
$$2k-2 = 2n-2$$

$$n = k$$

$$\text{Sum} = \frac{k}{2}(2k+1+4k-1)$$

$$= 3k^2$$

5(b)



Let $a_2, a_3, a_4 \dots$ be the lengths of the sides of the second, third, fourth rectangles, respectively

$$2(a_2)^2 = 50^2 \Rightarrow a_2 = \frac{50}{\sqrt{2}} \Rightarrow A_2 = \frac{50^2}{2}$$

$$2(a_3)^2 = (a_2)^2 = \frac{50^2}{2} \Rightarrow (a_3)^2 = \frac{50^2}{4} \Rightarrow A_3 = \frac{50^2}{4}$$

$$2(a_4)^2 = (a_3)^2 = \frac{50^2}{4} \Rightarrow (a_4)^2 = \frac{50^2}{8} \Rightarrow A_4 = \frac{50^2}{8}$$

Hence $A_1, A_2, \dots, A_n, \dots$ is a geometric series with first term 2500 and common ratio $\frac{1}{2}$

Or observe that area is half of previous area

$$A_n = 2500 \left(\frac{1}{2} \right)^{n-1} < 1$$

$$(n-1) \lg \left(\frac{1}{2} \right) < \lg \left(\frac{1}{2500} \right)$$

$$n-1 > \frac{\lg(1/2500)}{\lg(1/2)} = 11.3$$

$$n > 12.3$$

Least value of n is 13.

$$A_2 + A_4 + A_6 + A_8 + \dots = \frac{2500 \times \frac{1}{2}}{1 - \left(\frac{1}{2} \right)^2} = \frac{5000}{3} \text{ cm}^2 = 1666.7 \text{ cm}^2$$

6. Objective: Sampling Method (Stratified sampling)

6 strata (Pri 1, Pri 2, Pri 3, Pri 4, Pri 5, Pri, 6)

Stratum School Level (Primary)	Number of students (Sample Size)
1	$\frac{125}{1000} \times 400 = 50$
2	$\frac{200}{1000} \times 400 = 80$
3	$\frac{180}{1000} \times 400 = 72$
4	$\frac{150}{1000} \times 400 = 60$
5	$\frac{225}{1000} \times 400 = 90$
6	$\frac{120}{1000} \times 400 = 48$
Total	400

Conduct simple random sampling for each stratum to select the required sample.

7. Objective: Combinations & Permutations

Method 1

$$\text{No. of ways} = (7 - 1)! \times 10 \times {}^3C_2 \times 2! \times 2! = 86\,400$$

Method 2(Complementary Approach)

$$\text{No. of ways} = (7 - 1)! \times 10 \times (4! - 3!2!) = 86\,400$$

Number of ways the family seat together without restriction – number of ways the family seat together with Ann and Angela seating together)

Method 3 (Consider cases of which Ann and Angela not seated next to each other)

$$\text{No of ways} = (7 - 1)! \times 10 \times 2! \times 2! \times 3 = 86\,400$$

8. Objective: Hypothesis Testing

(ai) $H_0 : \mu = 8$ vs $H_1 : \mu > 8$

$X \sim$ the length, in millimeters, of a screw produced follows a normal distribution

(ii) Using GC, 1-var stats

$$\bar{x} = 8.0067 \text{ (5 s.f.)}, \quad s = 0.013229 \text{ (5 s.f.)}$$

$$\text{Test statistic, } t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = 1.5119 \text{ (5 s.f.)}$$

From GC, p -value = 0.084510 (5 s.f.)

The least level of significance that will result in the rejection of the null hypothesis is 8.46% (3 s.f.)

(b) $H_0 : \mu = 8$ vs $H_1 : \mu \neq 8$

Level of significance: 10%

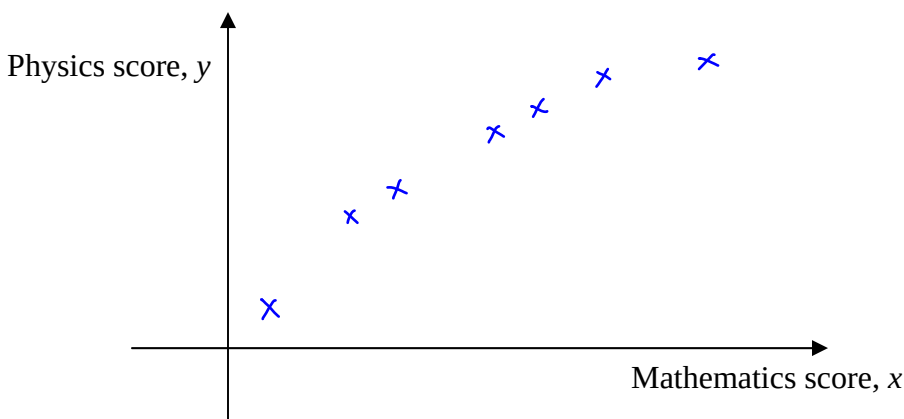
$$\text{Test statistic, } z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = 1.5462 \text{ (5 s.f.)}$$

From GC, p -value = 0.12207 (5 s.f.) > 0.1

Since the p -value is more than the level of significance, we do not reject H_0 . There is insufficient evidence at 10% level of significance that the machine is faulty.

9. **Objective: Use of scatter diagram and product moment correlation coefficient to determine suitability of model. Extrapolation and use of appropriate least square regression line.**

(i)



(ii) C: $y = a + \frac{b}{x}$ is definitely inappropriate as it is clear from the scatter diagram that as x

increases, y increases. [C suggests that as x increases, y decreases.]

Both Models A and B are possibilities.

Consider the product moment correlation coefficient for Model A = 0.9961

Product moment correlation coefficient for Model B = 0.9875

The most appropriate model is Model A.

(iii) Equation of regression line: $y = -71.370 + 36.695 \ln x$
 $\Rightarrow a = -71.4, b = 36.7$

- (iv) Sub $x = 10$,
 Estimated Physics score $y = -71.370 + 36.695 \ln 10 = 13.1$ marks
 As this is an extrapolation, the value obtained may not be valid as we cannot assume that there is a linear relationship outside the range of given values.
- (v) As we wish to estimate the value of x given a y value, the regression line of $\ln x$ on y should be used instead.

10. Objective: Normal Distribution

Given $M \sim$ mass of bottles. Then $M \sim N(200, 4.5^2)$.

$M + L \sim$ mass of filled bottles. Then $M + L \sim N(1200, 7.5^2)$.

- (i) $E(M + L) = E(M) + E(L) = 1200$
 $\Rightarrow E(L) = 1200 - 200 = 1000$
 $\text{Var}(M + L) = \text{Var}(M) + \text{Var}(L) = 7.5^2$
 $\Rightarrow \text{Var}(L) = 7.5^2 - 4.5^2 = 36$
 $\therefore$ standard deviation of $L = 6$ (Shown)
- (ii) $L \sim$ mass of fruit juice put into a bottle. Then $L \sim N(1000, 6^2)$.
 Consider $L_1 + L_2 - \frac{1}{2}(M + L) \sim N(1400, 86.0625)$
 $P(L_1 + L_2 - \frac{1}{2}(M + L) > 1395) = \text{normalcdf}(1395, 10^{99}, 1400, \sqrt{86.0625})$
 $= 0.705$ (3s.f.)

Given $L \sim N(1000, \sigma^2)$

$$P(|L - 1000| \leq 500) = 0.9$$

$$\Rightarrow P(|Z| \leq \frac{500}{\sigma}) = 0.9$$

$$\Rightarrow P(-\frac{500}{\sigma} \leq Z \leq \frac{500}{\sigma}) = 0.9$$

$$\Rightarrow P(Z \leq \frac{500}{\sigma}) = 0.95$$

$$\Rightarrow \frac{500}{\sigma} = \text{invNorm}(0.95, 0, 1) = 1.6449 \text{ (5s.f.)}$$

$$\therefore \sigma = 303.97 = 304 \text{ (3s.f.)}$$

11. Objective: Binomial, Poisson distributions , approximations

Let $K \sim$ number of Kampung handphones the salesman sells in a week.

Then $K \sim P_o(5.25)$

Let $T \sim$ number of Toki handphones the salesman sells in a week.

Then $T \sim P_o(8.75)$

Let $K + T \sim$ number of handphones the salesman sells in a week.

Then $K + T \sim P_o(14)$

$$\begin{aligned} \text{(i)} \quad & P(K + T > 12) \times P(K + T = 16) \\ &= [1 - P(K + T \leq 12)] \times P(K + T = 16) \\ &= [1 - \text{poissoncdf}(14, 12)] \times \text{poissonpdf}(14, 16) = 0.055530 = 0.0555 \text{ (3s.f.)} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad & P(T < 3 \mid K + T = 12) \\ &= P\left(\frac{T < 3 \text{ \& } K + T = 12}{K + T = 12}\right) \\ &= \frac{P(T = 0) \times P(K = 12) + P(T = 1) \times P(K = 11) + P(T = 2) \times P(K = 10)}{P(K + T = 12)} \\ &= \frac{\text{poissonpdf}(8.75, 0) \times \text{poissonpdf}(5.25, 12) + \text{poissonpdf}(8.75, 1) \times \text{poissonpdf}(5.25, 11) + \text{poissonpdf}(8.75, 2) \times \text{poissonpdf}(5.25, 10)}{\text{poissonpdf}(14, 12)} \\ &= 0.0015802 = 0.00158 \text{ (3s.f.)} \end{aligned}$$

(iii) Let $W \sim$ number of Kampung handphones sold by the salesman in a fourteen-week period.

Then $W \sim P_o(5.25 \times 14 = 73.5)$

Since $\mu = 73.5 > 10$, $\therefore W \sim N(73.5, 73.5)$ approximately.

$$\begin{aligned} P(W > 75) &= P(W > 75.5) \text{ (continuity correction)} \\ &= \text{normalcdf}(75.5, 10^{99}, 73.5, \sqrt{73.5}) = 0.40777 = 0.408 \text{ (3 s.f.)} \end{aligned}$$

$$P(K + T \geq 15) = 1 - P(K + T \leq 14) = 1 - \text{poissoncdf}(14, 14) = 0.42956 \text{ (5 s.f.)}$$

Let $V \sim$ number of weeks (out of 52) of which the salesman sells at least 15 handphones per week.

Then $V \sim B(52, 0.42956)$

Given $P(V > k) < 0.05$

$$\Rightarrow 1 - P(V \leq k) < 0.05$$

$$\Rightarrow P(V \leq k) > 0.95$$

Using GC, $P(V \leq 27) = 0.92532$, $P(V \leq 28) = 0.9572 > 0.95$

Therefore, the least value of $k = 28$.

12. Objective: CLT, Binomial , Poisson Distributions , approximation

(i) Let $X \sim$ number of words (out of 100) that the child pronounces wrongly.

Then $X \sim B(100, \frac{1}{20})$

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - \text{binomcdf}(100, \frac{1}{20}, 1) = 0.96292 = 0.963 \text{ (3 s.f.)}$$

$$\begin{aligned} \text{(ii) Required probability} &= P(X > 1) \times P(X = 1) \times P(X = 1) \times \frac{3!}{2!} \\ &= P(X \geq 2) \times P(X = 1) \times P(X = 1) \times \frac{3!}{2!} \\ &= [1 - \text{binomcdf}(100, \frac{1}{20}, 1)] \times \text{binompdf}(100, \frac{1}{20}, 1)^2 \times \frac{3!}{2!} \\ &= 0.0028049 = 0.00280 \text{ (3 s.f.)} \end{aligned}$$

(Alternative: use (i) ans, we get $0.0028050 = 0.00280$ (3 s.f.)

$$\text{(iii) } E(X) = 100 \times \frac{1}{20} = 5 \quad \text{Var}(X) = 100 \times \frac{1}{20} \times \frac{19}{20} = 4.75$$

Let $\bar{X} \sim$ average number of words per book that the child pronounces wrongly

Since $n = 50$ is large, by Central Limit Theorem, $\bar{X} \sim N(5, \frac{4.75}{50})$ approximately.

$$P(4 \leq \bar{X} \leq 7) = \text{normalcdf}(4, 7, 5, \sqrt{\frac{4.75}{50}}) = 0.99941 = 0.999 \text{ (3 s.f.)}$$

$$\text{(iv) } P(X < 2) = 1 - 0.96292 \text{ (ans from (i))} = 0.03708$$

$$\text{Alternative, } P(X < 2) = P(X \leq 1) = \text{binomcdf}(100, \frac{1}{20}, 1) = 0.037081$$

Let $Y \sim$ number of books (out of 50) of which the child pronounces fewer than 2 words wrongly.

Then $Y \sim B(50, 0.03708)$. (or $Y \sim B(50, 0.037081)$)

Since $n = 50$ is large, $np = 50 \times 0.03708 = 1.854$ (or 1.85405)

$\therefore Y \sim P_0(1.854)$ approximately.

$$P(Y < 4) = P(Y \leq 3) = \text{poissoncdf}(1.854, 3) = 0.882 \text{ (3 s.f.)}$$

Note : More than 46 books of which the child pronounces at least 2 words wrongly per book is equivalent to fewer than 4 books of which he pronounces fewer than 2 words wrongly per book. 'Complement' case must be used if not $np > 5$, $n(1-p) < 5$